if the receiver gets closer to than , the defender at cannot intercept (Fig. ). If , it is part . If , let the receiver be at on .

His time to reach is

Defender arrival time at :

Fig.

The condition for avoiding interception is

Thus

where is given by .

Method 2: the defender and ball reach at the same time, so

The no-interception condition and equation give

Again is given by . Equations and clearly agree.

2.

In a chosen frame, satellite motion gives its coordinates at every instant. A signal emitted at reaches ground station at .

The signal travels at light speed , hence

are known, premeasured station coordinates; its clock measures .

The signal supplies , also known. Equation thus has three unknowns, , requiring three independent equations: at least three ground stations.

These are for .

Earth centre and stations form an isosceles triangle; the equal sides are . Thus